C17 Thermodynamics – Knowledge Organiser

AQA A-level Chemistry • Set 1: Born–Haber Definitions **Paper 1**

#	Question	Answer
1	Enthalpy of atomisation	Enthalpy change when 1 mol of gaseous atoms forms from the element in its standard state
2	First ionisation energy	Enthalpy change to remove 1 mol of electrons from 1 mol of gaseous atoms
3	First electron affinity	Enthalpy change when 1 mol of gaseous atoms each gain 1 electron to form 1 mol of gaseous 1− ions
4	Second electron affinity	Enthalpy change when 1 mol of gaseous 1− ions each gain 1 electron to form 1 mol of gaseous 2− ions
5	Sign of first electron affinity	Negative
6	Sign of second electron affinity	Positive
7	Lattice enthalpy of formation	Enthalpy change when 1 mol of ionic solid forms from its gaseous ions
8	Lattice enthalpy of dissociation	Enthalpy change when 1 mol of ionic solid separates into its gaseous ions
9	Enthalpy of hydration	Enthalpy change when 1 mol of gaseous ions dissolves in water
10	Enthalpy of solution	Enthalpy change when 1 mol of solute dissolves completely in water

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AQA A-level Chemistry • Set 2: Lattice Enthalpy and Entropy **Paper 1**

#	Question	Answer
1	Bond dissociation enthalpy	Enthalpy change to break 1 mol of a covalent bond, all species gaseous
2	Equation for enthalpy of solution	$\Delta_{sol}H = \Delta_{L,diss}H + \Sigma\Delta_{hyd}H$
3	Sign of enthalpy of hydration	Negative
4	Two assumptions of the perfect ionic model	Ions are perfect spheres; bonding is purely ionic
5	Experimental lattice enthalpy larger than theoretical means	Covalent character
6	Two factors giving a larger lattice enthalpy	Smaller ions, higher charges
7	Entropy (S)	Measure of disorder
8	Units of entropy	$J\ K^{-1}\ mol^{-1}$
9	Order of entropy for states	Gas > liquid > solid
10	Entropy of a perfect crystal at 0 K	0

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AQA A-level Chemistry • Set 3: Entropy and Gibbs Free Energy **Paper 1**

#	Question	Answer
1	Equation for ΔS	ΣS products − ΣS reactants
2	Gibbs free energy equation	$\Delta G = \Delta H - T\Delta S$
3	Condition for a reaction to be feasible	$\Delta G \leq 0$
4	Units of ΔG	kJ mol^{-1}
5	Convert ΔS to use with ΔH in kJ	÷ 1000
6	Temperature at which a reaction becomes feasible	$T = \Delta H \div \Delta S$
7	Gradient of a ΔG against T graph	$-\Delta S$
8	y-intercept of a ΔG against T graph	ΔH
9	Sign of ΔS when moles of gas increase	Positive
10	Why a feasible reaction may not happen	High activation energy